

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459844.9175 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1672 +/- 0.0097
Transit depth (Rp/Rs)^2	2.8 +/- 0.32 [%]
Orbital Inclination (inc)	84.03 +/- 0.95
Airmass coefficient 1 (a1)	1.0142 +/- 0.0095
Airmass coefficient 2 (a2)	-0.009 +/- 0.041
Scatter in the residuals of the lightcurve fit is	0.94 %
Best Comparison Star	#3 - [391, 355]
Optimal Aperture	4.07
Optimal Annulus	12.09
Transit Duration (day)	0.121 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1672 $\pm$ 0.0097 vs 0.1489 (deviation 1.76 σ)
Duration fit vs published	0.121 d vs 0.1304 d (deviation 2.1 σ)
Mid-time O-C	-0.1 min
Depth SNR	16.1
Residual scatter	0.94 %

Light curve

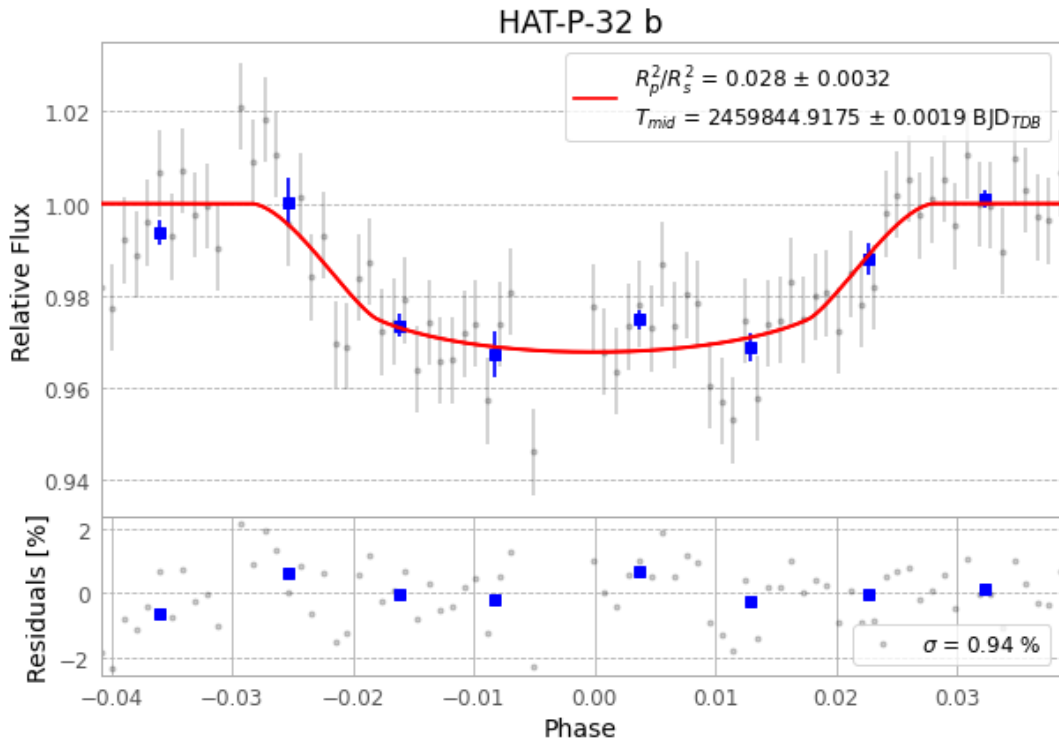


Figure 1 — FinalLightCurve_HAT-P-32 b_2022-09-22.png (SHA-256 0ac978c38fb6c3bd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [400, 298], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c5d530a071d29217...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

83 FITS frames (55 MB), HATP-32220922074816.FITS → HATP-32220922115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-220922.log (2c42a6759cc3...), FinalLightCurve_HAT-P-32 b_2022-09-22.png (0ac978c38fb6...), FinalLightCurve_HAT-P-32 b_2022-09-22.csv (bd89ccec8db7...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 07:13Z.